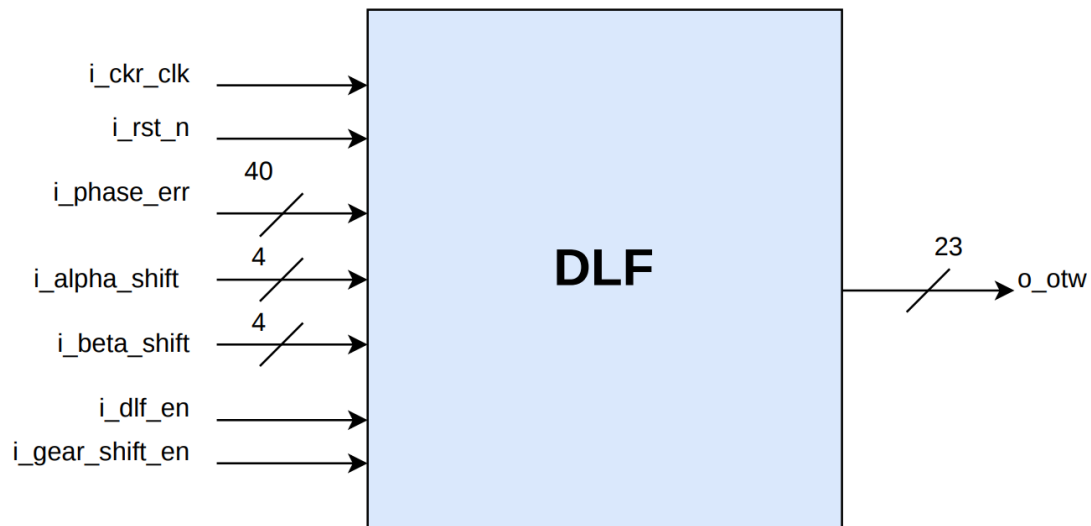
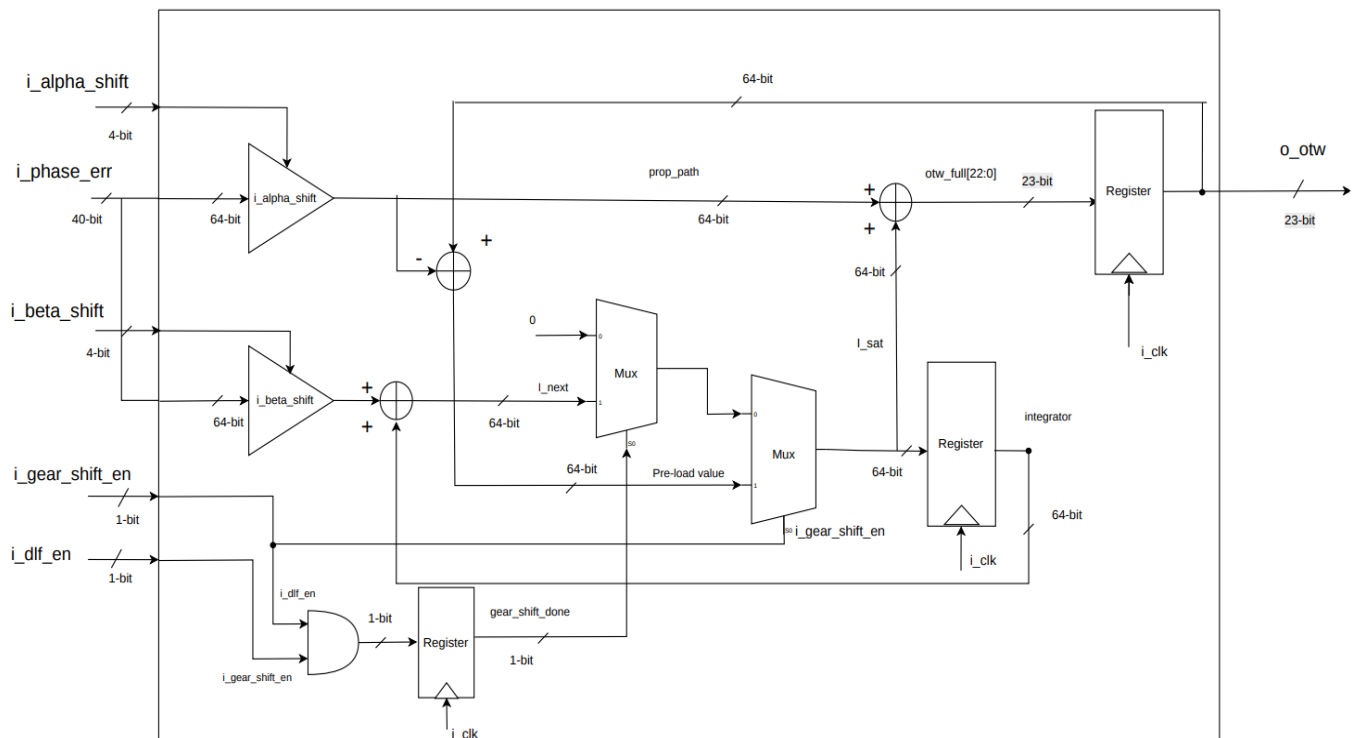


Digital loop filter specifications

Abstraction:



Block diagram:



Signal Descriptions:

Signal	Port type	Width	Description
i_ckr_clk	Input	1	Input clock ckr (crystal clock over sampled by dco)
i_rst_n	Input	1	Active low asynchronous reset.
i_dlf_en	Input	1	Active high digital loop filter enable.
i_gear_shift_en	Input	1	Gear shift control signal. High at the start of normal tracking mode.
i_alpha_shift	Input	4	Proportional gain as shift right value.
i_beta_shift	Input	4	Integral gain as shift right value.
i_phase_err	Input	40	Input phase error of TDC.
o_otw	Output	23	DCO tuning word.